NT-C-002 | Java Full Stack | Course Knowledge Profile

Identity, purpose, prerequisites, scope and boundaries

Category	Programming & Full Stack
Target learners	Learners targeting enterprise Java and web-application roles.
Prerequisites	Basic logic; prior programming is helpful.
Approved duration	Approved working range: 20-24 weeks / 240-300 guided hours.
Final outcome	Develop and deploy a Java/Spring Boot application with REST services, persistence, security and a responsive front end.
Assessment profile	Programming/Platform

Approved tools / platforms

Java, Spring Boot, Spring MVC, Spring Data JPA, Hibernate, SQL, HTML, CSS, JavaScript, React, Git, Maven/Gradle.

Knowledge boundaries

Include	Quality rule	Exclude / escalate
The eight approved modules, four sections, named tools, projects and mapped entry-level roles.	Prefer runnable or demonstrable outputs, clear input validation, error handling, tests and version control.	Do not treat copied code or syntax recall as proof of implementation skill.
Current product/platform procedures only when version and date are known.	Use primary documentation and record the source version.	Do not invent current commands, prices, tax rates, policies or certification status.
Evidence-based career guidance.	Cite tests, projects, capstone, viva and verified resume evidence.	No guaranteed job, placement, salary, interview or employer claim.

LLM answer pattern: identify the course and version, answer from approved records, cite record IDs, distinguish verified facts from guidance, and state any missing evidence.

NT-C-002 | Java Full Stack | NT-C-002-S01 Deep Knowledge

Modules 1-2 | objectives, concepts, evidence and remediation

Section competency	Syntax, control flow, methods, arrays; OOP, collections, exceptions, streams
Completion evidence	Console programs; Tested Java application
Section weight	20% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 code/design scenarios (30) + 2 practical tasks (40)

Module 1 - Java foundations

Objective: Syntax, control flow, methods, arrays
Observable evidence: Console programs

Concept	Approved knowledge statement	Application behaviour	Evidence
Syntax	The formal rules used to write valid instructions in a programming language. In Java Full Stack, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when syntax is appropriate; apply it in a Java Full Stack task; verify the result.	Console programs
control flow	The order in which conditions, loops and branches execute. In Java Full Stack, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when control flow is appropriate; apply it in a Java Full Stack task; verify the result.	Console programs
methods	A defined competency within Java Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when methods is appropriate; apply it in a Java Full Stack task; verify the result.	Console programs
arrays	A defined competency within Java Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when arrays is appropriate; apply it in a Java Full Stack task; verify the result.	Console programs

Module 2 - Core Java

Objective: OOP, collections, exceptions, streams
Observable evidence: Tested Java application

Concept	Approved knowledge statement	Application behaviour	Evidence
OOP	A design approach that organises behaviour and data around objects and clear responsibilities. In Java Full Stack, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when oop is appropriate; apply it in a Java Full Stack task; verify the result.	Tested Java application
collections	A defined competency within Java Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when collections is appropriate; apply it in a Java Full Stack task; verify the result.	Tested Java application
exceptions	A defined competency within Java Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when exceptions is appropriate; apply it in a Java Full Stack task; verify the result.	Tested Java application
streams	A defined competency within Java Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when streams is appropriate; apply it in a Java Full Stack task; verify the result.	Tested Java application

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Console programs; Tested Java application, return to Modules 1-2, complete one guided practice and then use a new equivalent test form.

NT-C-002 | Java Full Stack | NT-C-002-S02 Deep Knowledge

Modules 3-4 | objectives, concepts, evidence and remediation

Section competency	SQL, schema design, JDBC; HTML, CSS, JavaScript, React
Completion evidence	Database-backed exercise; Responsive UI
Section weight	25% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 code/design scenarios (30) + 2 practical tasks (40)

Module 3 - Database access

Objective: SQL, schema design, JDBC
Observable evidence: Database-backed exercise

Concept	Approved knowledge statement	Application behaviour	Evidence
SQL	A language for defining, querying and controlling relational data. In Java Full Stack, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when sql is appropriate; apply it in a Java Full Stack task; verify the result.	Database-backed exercise
schema design	A defined competency within Java Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when schema design is appropriate; apply it in a Java Full Stack task; verify the result.	Database-backed exercise
JDBC	A defined competency within Java Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when jdbc is appropriate; apply it in a Java Full Stack task; verify the result.	Database-backed exercise

Module 4 - Front-end skills

Objective: HTML, CSS, JavaScript, React
Observable evidence: Responsive UI

Concept	Approved knowledge statement	Application behaviour	Evidence
HTML	A defined competency within Java Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when html is appropriate; apply it in a Java Full Stack task; verify the result.	Responsive UI
CSS	A defined competency within Java Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when css is appropriate; apply it in a Java Full Stack task; verify the result.	Responsive UI
JavaScript	A defined competency within Java Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when javascript is appropriate; apply it in a Java Full Stack task; verify the result.	Responsive UI
React	A defined competency within Java Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when react is appropriate; apply it in a Java Full Stack task; verify the result.	Responsive UI

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Database-backed exercise; Responsive UI, return to Modules 3-4, complete one guided practice and then use a new equivalent test form.

NT-C-002 | Java Full Stack | NT-C-002-S03 Deep Knowledge

Modules 5-6 | objectives, concepts, evidence and remediation

Section competency	Project structure, dependency injection, configuration; Spring MVC, REST, JPA/Hibernate
Completion evidence	Spring Boot service; CRUD REST API
Section weight	25% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 code/design scenarios (30) + 2 practical tasks (40)

Module 5 - Spring Boot

Objective: Project structure, dependency injection, configuration
Observable evidence: Spring Boot service

Concept	Approved knowledge statement	Application behaviour	Evidence
Project structure	A defined competency within Java Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when project structure is appropriate; apply it in a Java Full Stack task; verify the result.	Spring Boot service
dependency injection	A defined competency within Java Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when dependency injection is appropriate; apply it in a Java Full Stack task; verify the result.	Spring Boot service
configuration	A defined competency within Java Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when configuration is appropriate; apply it in a Java Full Stack task; verify the result.	Spring Boot service

Module 6 - REST & persistence

Objective: Spring MVC, REST, JPA/Hibernate
Observable evidence: CRUD REST API

Concept	Approved knowledge statement	Application behaviour	Evidence
Spring MVC	A defined competency within Java Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when spring mvc is appropriate; apply it in a Java Full Stack task; verify the result.	CRUD REST API
REST	An http-oriented interface style using resources, methods and predictable responses. In Java Full Stack, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when rest is appropriate; apply it in a Java Full Stack task; verify the result.	CRUD REST API
JPA/Hibernate	A defined competency within Java Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when jpa/hibernate is appropriate; apply it in a Java Full Stack task; verify the result.	CRUD REST API

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Spring Boot service; CRUD REST API, return to Modules 5-6, complete one guided practice and then use a new equivalent test form.

NT-C-002 | Java Full Stack | NT-C-002-S04 Deep Knowledge

Modules 7-8 | objectives, concepts, evidence and remediation

Section competency	Validation, Spring Security, JUnit, API testing; Front/back integration, packaging, deployment
Completion evidence	Secured tested service; Deployed capstone
Section weight	30% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 code/design scenarios (30) + 2 practical tasks (40)

Module 7 - Security & testing

Objective: Validation, Spring Security, JUnit, API testing
Observable evidence: Secured tested service

Concept	Approved knowledge statement	Application behaviour	Evidence
Validation	Checking input or output against defined rules before trusting it. In Java Full Stack, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when validation is appropriate; apply it in a Java Full Stack task; verify the result.	Secured tested service
Spring Security	A defined competency within Java Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when spring security is appropriate; apply it in a Java Full Stack task; verify the result.	Secured tested service
JUnit	A defined competency within Java Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when junit is appropriate; apply it in a Java Full Stack task; verify the result.	Secured tested service
API testing	Systematic verification that expected behaviour and important failure cases work correctly. In Java Full Stack, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when api testing is appropriate; apply it in a Java Full Stack task; verify the result.	Secured tested service

Module 8 - Integration & deployment

Objective: Front/back integration, packaging, deployment
Observable evidence: Deployed capstone

Concept	Approved knowledge statement	Application behaviour	Evidence
Front/back integration	A defined competency within Java Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when front/back integration is appropriate; apply it in a Java Full Stack task; verify the result.	Deployed capstone
packaging	A defined competency within Java Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when packaging is appropriate; apply it in a Java Full Stack task; verify the result.	Deployed capstone
deployment	A defined competency within Java Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when deployment is appropriate; apply it in a Java Full Stack task; verify the result.	Deployed capstone

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Secured tested service; Deployed capstone, return to Modules 7-8, complete one guided practice and then use a new equivalent test form.

NT-C-002 | Java Full Stack | Skill Ontology & Dependency Map

Atomic skills, learning order and evidence strength

Skill ID	Competency	Depends on	Evidence	Type
NT-C-002-SK01	Java foundations	Prerequisite foundation	Console programs	Critical
NT-C-002-SK02	Core Java	M01 plus earlier foundations	Tested Java application	Critical
NT-C-002-SK03	Database access	M02 plus earlier foundations	Database-backed exercise	Supporting
NT-C-002-SK04	Front-end skills	M03 plus earlier foundations	Responsive UI	Supporting
NT-C-002-SK05	Spring Boot	M04 plus earlier foundations	Spring Boot service	Supporting
NT-C-002-SK06	REST & persistence	M05 plus earlier foundations	CRUD REST API	Critical
NT-C-002-SK07	Security & testing	M06 plus earlier foundations	Secured tested service	Supporting
NT-C-002-SK08	Integration & deployment	M07 plus earlier foundations	Deployed capstone	Critical

Evidence-strength hierarchy

Strength	Evidence source	Use
High	Trainer-observed practical, viva, working source/project files and reproducible output.	May support verified mastery when rubric threshold is met.
Medium	Scored mock tests, reviewed assignments and documented portfolio artifacts.	Supports mastery with corroboration.
Low	Resume claim, self-report, attendance or video completion without demonstration.	May trigger verification; cannot alone support Apply Now.

CareerTours must not treat attendance, time spent or content opened as skill mastery.

NT-C-002 | Java Full Stack | Assessment Knowledge & Sample Items

Non-live examples for LLM training and evaluation

Sample ID	Type	Question	Approved answer / rubric anchor
NT-C-002-S01-EX01	Evidence/application	What evidence best demonstrates the combined competency of Java foundations and Core Java?	The learner should provide Console programs; Tested Java application and explain how the work applies Syntax, control flow, methods, arrays; OOP, collections, exceptions, streams. Recall alone is insufficient.
NT-C-002-S01-EX02	Scenario/remediation	A learner reports completing Core Java but cannot produce Tested Java application. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-002-S01, request Tested Java application, and allow a new equivalent test only after remediation.
NT-C-002-S02-EX01	Evidence/application	What evidence best demonstrates the combined competency of Database access and Front-end skills?	The learner should provide Database-backed exercise; Responsive UI and explain how the work applies SQL, schema design, JDBC; HTML, CSS, JavaScript, React. Recall alone is insufficient.
NT-C-002-S02-EX02	Scenario/remediation	A learner reports completing Front-end skills but cannot produce Responsive UI. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-002-S02, request Responsive UI, and allow a new equivalent test only after remediation.
NT-C-002-S03-EX01	Evidence/application	What evidence best demonstrates the combined competency of Spring Boot and REST & persistence?	The learner should provide Spring Boot service; CRUD REST API and explain how the work applies Project structure, dependency injection, configuration; Spring MVC, REST, JPA/Hibernate. Recall alone is insufficient.
NT-C-002-S03-EX02	Scenario/remediation	A learner reports completing REST & persistence but cannot produce CRUD REST API. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-002-S03, request CRUD REST API, and allow a new equivalent test only after remediation.
NT-C-002-S04-EX01	Evidence/application	What evidence best demonstrates the combined competency of Security & testing and Integration & deployment?	The learner should provide Secured tested service; Deployed capstone and explain how the work applies Validation, Spring Security, JUnit, API testing; Front/back integration, packaging, deployment. Recall alone is insufficient.
NT-C-002-S04-EX02	Scenario/remediation	A learner reports completing Integration & deployment but cannot produce Deployed capstone. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-002-S04, request Deployed capstone, and allow a new equivalent test only after remediation.

Live bank rule: Generate at least three times the live quantity for each section using the approved Programming/Platform blueprint. Human-review every scored item before initial production release.

NT-C-002 | Java Full Stack | Labs, Projects & Scoring Rubrics

Practical proof of learning and ownership

Level	Approved project	Skills evidenced	Required deliverables
Mini 1	Inventory manager	Java, collections, SQL	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Mini 2	Support-ticket REST API	Spring Boot, JPA, validation	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Capstone	Admissions and fee-management portal	Security, React, APIs, deployment	Brief; working/source file or reproducible environment; output; validation notes; reflection.

Standard project rubric

Dimension	Weight	What earns credit
Problem framing	15%	Clear objective, user/business need, assumptions and success criteria.
Technical/design accuracy	25%	Correct methods, appropriate tools and sound decisions.
Completeness	15%	Required features, assets, data or workflow are present.
Testing and validation	15%	Important normal, edge, quality or failure cases are checked.
Evidence and reproducibility	15%	Source/working files, setup, inputs, outputs and version information are available.
Documentation	5%	Concise structure, instructions and limitations.
Viva / ownership	10%	Learner can explain decisions, demonstrate work and respond to a change request.

Prefer runnable or demonstrable outputs, clear input validation, error handling, tests and version control.

NT-C-002 | Java Full Stack | Career Roles, Evidence & Gap Actions

Explainable top-three occupation matching

Rank	Occupation	Default weighted skills	Minimum evidence	Gap action
1	Junior Java Developer	Core Java 25% [critical]; OOP 25% [critical]; SQL 25%; Git 25%	Tested Java repository	If a critical skill is below 60 or tested java repository is missing, return Improve First or Needs Evidence with the linked module/project.
2	Spring Boot Developer - Junior	REST 25% [critical]; JPA 25% [critical]; security 25%; testing 25%	Deployed API	If a critical skill is below 60 or deployed api is missing, return Improve First or Needs Evidence with the linked module/project.
3	Full Stack Java Developer	React 34% [critical]; Spring Boot 33% [critical]; database 33%	End-to-end capstone	If a critical skill is below 60 or end-to-end capstone is missing, return Improve First or Needs Evidence with the linked module/project.

Course readiness	45% section mocks + 20% mini projects + 25% capstone + 10% viva.
Role readiness	45% role-relevant mastery + 25% projects/capstone + 20% aligned mocks + 10% verified resume evidence.
Evidence confidence	Show separately; Apply Now requires at least 70.
Resume extraction	Use only relevant skills, projects, education, experience and portfolio evidence; minimise personal data before external model processing.

A student may be Apply Now for one mapped occupation and Improve First for another. Return independent evidence and gaps for each role.

NT-C-002 | Java Full Stack | FAQs, Misconceptions & LLM Response Pack

Course-specific retrieval and response patterns

Approved FAQ	Answer
Who is Java Full Stack for?	Learners targeting enterprise Java and web-application roles.
What should I know before starting?	Basic logic; prior programming is helpful.
How long is the approved learning journey?	Approved working range: 20-24 weeks / 240-300 guided hours.
What will I be able to demonstrate?	Develop and deploy a Java/Spring Boot application with REST services, persistence, security and a responsive front end.
Does completion guarantee a job?	No. Completion and role readiness are evidence-based guidance; employment, placement, salary and employer selection are not guaranteed.
What if I fail a section?	CareerTours identifies the failed skill tags and module, assigns targeted practice, and permits an equivalent retest after remediation.

Misconception	Approved correction
Attendance proves mastery	False - require tests and observable practical/project evidence.
One total score proves every role fit	False - calculate each occupation using its weighted skills and evidence.
The LLM can add missing details	False - retrieve approved records or state that evidence is insufficient.
Course completion guarantees employment	False - never claim guaranteed job, placement, salary or interview.
Java Full Stack knowledge never changes	False - version-sensitive tools, laws, taxes, security and platform procedures require review.

Retrieval metadata

```
course_code=NT-C-002; corpus_version=3.0;
record_types=course_profile,module,section,skill,assessment_sample,project,occupation_mapping,faq,misconception;
approval_status=approved; effective_date=2026-08-12
```